

Asian Options (2. The PDE)

Strongly Path-dependent Options

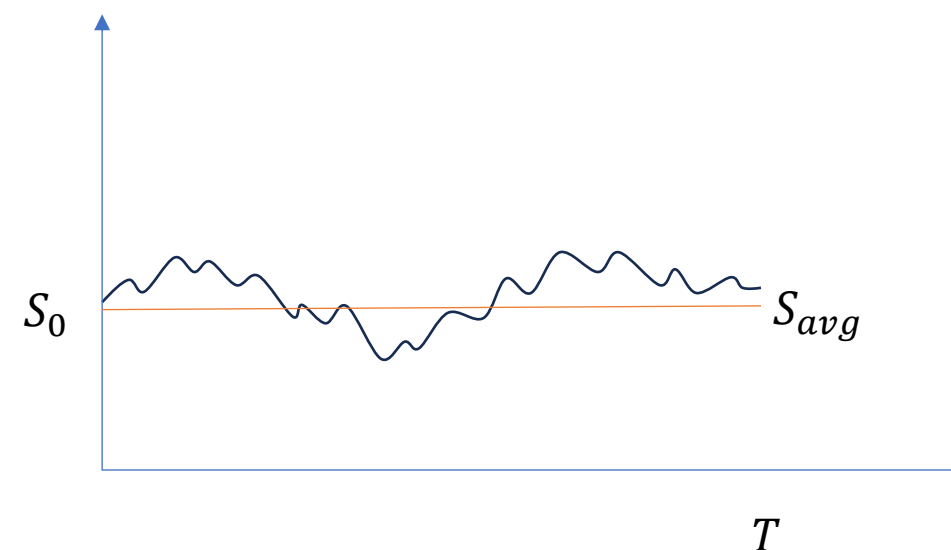
Asian options are strongly path dependent (recall Barrier options are weakly path dependent) as they depend on the entire path of the stock price. Path dependent options are typically solved using numerical methods. For low dimensional problems, we can use finite difference but as the dimension gets higher, we resort to Monte Carlo Simulation

numerical methods.

- Monte Carlo Simulation ✓
- Finite Difference Method ✓ ~~CPDE~~

⑤
(s_1, s_2)

To use finite difference method, we need to develop the PDE for the price



The state variable I

Let's imagine a contract has a payoff at time T that is a function of the path taken by the asset from time 0 to T . Let's assume the path-dependent quantity is expressed as the integral of some function

$$I \rightarrow I(T) = \int_0^T f(S, \tau) d\tau \quad \begin{aligned} I(T) &= \int_0^T f(S, u) du \\ I(0) &= \int_0^0 f(S, u) du \end{aligned}$$

Most of the path dependent options can be written in the above form with a suitable choice of $f(S, t)$

The payoff at time T is given as $P(S, I)$ (S_T, I_T) (S, I)

We can see that the value of the option is not only a function of S but also I . We will need to treat I as a new state variable. The SDE satisfied by I is

$$dI = f(S, t)dt \quad \begin{aligned} \frac{dI}{dt} &= f(S, t) \\ dI &= \int_0^t f(S, u) du \end{aligned}$$

$$dV = ()dt + ()dw$$

Examples

Depending on the contract $f(S, t)$ takes various forms

For Asian Option with arithmetic averaging

$$I(t) = \int_0^t \underline{S} d\tau$$

$$P(\underline{S}, I) = \max \left\{ \frac{I}{T} - K, 0 \right\}$$

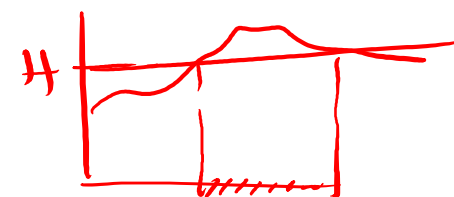
For Asian Option with geometric averaging

$$I(t) = \int_0^t \log(S) d\tau$$

$$P(S, I) = \max \left\{ \exp \left(\frac{I}{T} \right) - K, 0 \right\}$$

For a contract that pays an amount proportional to the time it spent above a certain level

$$I(t) = \int_0^t (1_{S>H}) d\tau$$



$$\max \left(S - \frac{I}{T}, 0 \right)$$

and so on...

The PDE (continuous sampling)

The pricing PDE can be derived for the path-dependent contract by using the same delta-hedging argument we used to derive Black-Scholes PDE. The only additional consideration is that we must treat the value of the option as a function of 3 variables

Consider the portfolio $\Pi = V(S, I, t) - \Delta S$

$$\Delta = \frac{\partial V}{\partial S}$$

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (dS)^2 + \frac{\partial V}{\partial I} dI + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma^2 dt$$

The change in value of the portfolio $d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \frac{\partial V}{\partial I} dI + \left(\frac{\partial V}{\partial S} - \Delta \right) dS$

Using $\Delta = \frac{\partial V}{\partial S}$, the portfolio becomes risk-free and it needs to grow at risk-free rate r . This leads to the pricing equation

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + \underbrace{f(S, t) \frac{\partial V}{\partial I}}_{\text{(additional)}} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$\frac{\partial V}{\partial t} \cdot f(S, t) dt$$

Asian Option PDE (continuous sampling)

For a specific contract, the PDE needs to be adjusted for the appropriate I and need to be solved using the right boundary conditions. For Arithmetic Asian Option,

$$I(t) = \int_0^t \underline{S} d\tau$$

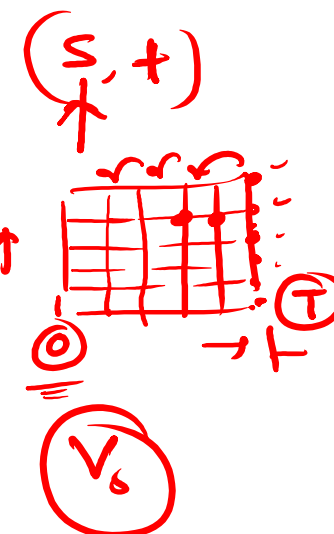
$$\frac{I}{T}$$

The pricing PDE is

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + \underline{S} \frac{\partial V}{\partial I} + rS \frac{\partial V}{\partial S} - rV = 0$$

discretize the PDE

PDE



$$f \frac{\partial V}{\partial I}$$

$$f \rightarrow S$$

$$f \rightarrow (\log S)$$

$$[S, I], t$$

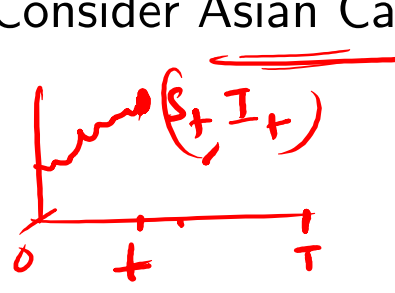
$$\Delta V, \Delta t, \Delta S$$

Similarity Reduction

The PDE we developed has 2 state variables (S, I). We can drop one dimension by using suitable variable transformation called as similarity reduction.

How to create the suitable transformation ? The payoff function gives some idea.

Consider Asian Call with floating strike for example



$$\boxed{P(S_T, I_T)} = \max \left\{ \underline{S_T} - \underline{\frac{I_T}{T}}, 0 \right\} = \underline{(S_T)} \max \left\{ 1 - \frac{1}{T} \left(\frac{I_T}{S_T} \right), 0 \right\}$$

Define $\underline{x} = \frac{I_t}{S_t}$ $\underline{V(S_t, I_t, t)} = \underline{(S_t)} \underline{H(x, t)}$, for the final payoff $\underline{H(x, T)} = \max \left\{ 1 - \frac{1}{T} \left(\frac{I_T}{S_T} \right), 0 \right\}$

Let's derive the PDE with our new variables x and $\underline{(H)}$

Similarity Reduction

After simplification, the PDE reduces to

$$\checkmark \quad \frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 H}{\partial x^2} + (1 - rx) \frac{\partial H}{\partial x} = 0$$

In this equation, we have only one state variable x and the PDE is easily solved using 1D FD scheme.

$$x = (I/S)$$

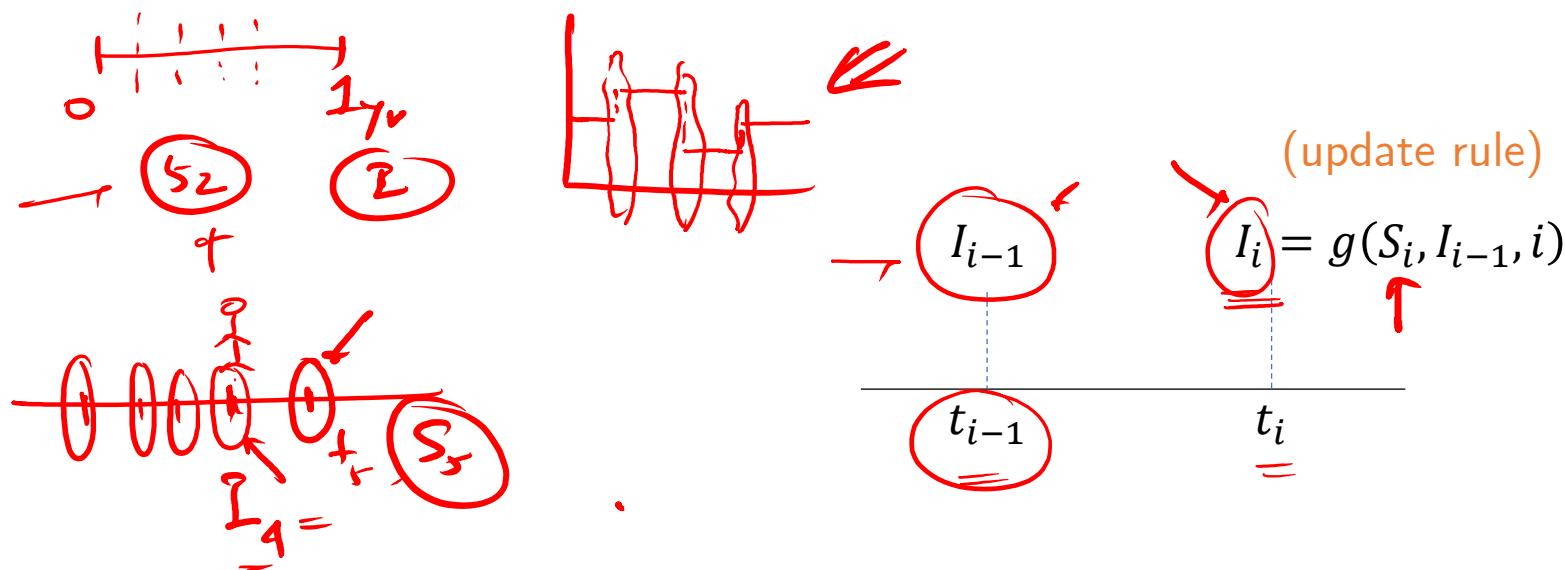
$$I(t) = \int_0^t \log S_u du$$

Exercise – For geometric averaging fixed strike Asian, with pay off $\max\{e^{I_T/T} - K, 0\}$, where $I(t) = \int_0^t \log(S) d\tau$, by considering the change of variable $x = \frac{I_t + (T-t) \log(S_t)}{T}$ the PDE reduces to

$$\rightarrow \frac{\partial H}{\partial t} + \frac{1}{2} \left(\frac{\sigma(T-t)}{T} \right)^2 \frac{\partial^2 H}{\partial x^2} + \left(r - \frac{1}{2} \sigma^2 \right) \left(\frac{T-t}{T} \right) \frac{\partial H}{\partial x} - rH = 0$$

Discrete Sampling

Practically speaking the path-dependent quantities are sampled in discrete fashion looking at closing prices. The frequency of sampling maybe one day, one week or one month... The path dependent quantity is measured on sampling dates t_i and takes a value I_t for $t_i \leq t \leq t_{i+1}$. At the sampling date t_i the quantity I_{i-1} can be updated using the following rule



What is the update rule for a discretely sampled Asian option with arithmetic averaging ?

Asian Option : (Update Rule)

For a discretely sampled Asian option, let's consider the following quantities

M

$$\underline{I_M} = \sum_{i=1}^M \underline{S_i}$$
 (sum)

$$\underline{A_M} = \frac{\underline{I_M}}{\underline{M}} = \frac{1}{M} \sum_{i=1}^M S_i$$
 (average)

P(S, A) = max{ $\underline{A_M}$ - S, 0}
 (payoff)

I, A

$$\underline{A_i} = \frac{1}{i} \underline{S_i} + \frac{i-1}{i} \underline{A_{i-1}}$$

What is the update rule for Lookback option ?

$$\underline{I_i} = \max\{S_i, \underline{I_{i-1}}\}$$

$$\underline{\max(S_{\max} - k, 0)}$$

Pricing equation (discrete sampling)

Since I does not change continuously, the SDE for I is degenerate. This is because I only changes (jumps) on sampling dates. V is still a function of three variables, but I is just used as a parameter.

$$I(t) = \int_0^t S_u du \quad \leftarrow \quad \left(\frac{\partial V}{\partial I} dI \right) \quad \boxed{dI = 0}$$

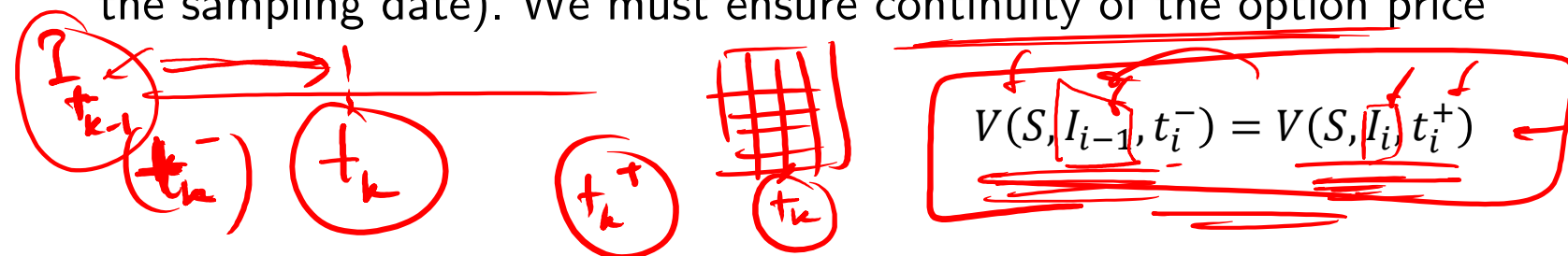
$$I(t) = \int_0^t f(S_u, u) du$$

$$dI = f(S_t) dt$$

The pricing equation is therefore the basic Black-Scholes equation

$$I_{t_k^+} = I_{t_k^-} + S_{t_k} \Rightarrow \boxed{\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0}$$

But this equation is oblivious to path dependencies. Although I jumps on sampling dates, the price of the option is continuous across a sampling date. (we become increasingly sure about I as we approach the sampling date). We must ensure continuity of the option price



$$V(S, I_{i-1}, t_i^-) = V(S, I_i, t_i^+)$$

Pricing equation (discrete sampling)

For **discrete sampling**, the underlying SDE is still plain Black-Scholes between sampling dates. The path dependence enters only through the **sampling updates**, so the structure is:

between fixing dates: standard Black-Scholes evolution

at fixing dates: enforce a jump condition

More on this later ...